```
PS C:\Users\homes> arp -a

Interface: 10.87.222.52 --- 0x3
  Internet Address      Physical Address      Type
  10.87.222.194         3a-10-be-39-a4-0b     dynamic
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static

Interface: 192.168.56.1 --- 0x8
  Internet Address      Physical Address      Type
  192.168.56.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static
PS C:\Users\homes> hostname
HomeshKumar31
PS C:\Users\homes> ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : HomeshKumar31
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Mixed
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Ethernet adapter Ethernet 3:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : VirtualBox Host-Only Ethernet Adapter
   Physical Address. . . . . . . . . : 0A-00-27-00-00-08
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::e0ec:779a:3248:ff27%8(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.56.1(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
```

```
PS C:\Users\homes> nbtstat

Displays protocol statistics and current TCP/IP connections using NBT
(NetBIOS over TCP/IP).

NBTSTAT [ [-a RemoteName] [-A IP address] [-c] [-n]
        [-r] [-R] [-RR] [-s] [-S] [interval] ]

  -a   (adapter status) Lists the remote machine's name table given its name
  -A   (Adapter status) Lists the remote machine's name table given its
                        IP address.
  -c   (cache)          Lists NBT's cache of remote [machine] names and their IP addresses
  -n   (names)          Lists local NetBIOS names.
  -r   (resolved)       Lists names resolved by broadcast and via WINS
  -R   (Reload)         Purges and reloads the remote cache name table
  -s   (Sessions)       Lists sessions table with the destination IP addresses
  -S   (sessions)       Lists sessions table converting destination IP
                        addresses to computer NETBIOS names.
  -RR  (ReleaseRefresh) Sends Name Release packets to WINS and then, starts Refresh

  RemoteName   Remote host machine name.
  IP address   Dotted decimal representation of the IP address.
  interval     Redisplays selected statistics, pausing interval seconds
               between each display. Press Ctrl-C to stop redisplaying
               statistics.

PS C:\Users\homes> netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    10.87.222.52:49419     4.213.25.242:https     ESTABLISHED
  TCP    10.87.222.52:49557     49.44.131.96:https     ESTABLISHED
  TCP    10.87.222.52:50347     150.171.27.11:https    TIME_WAIT
PS C:\Users\homes> nslookup www.google.com
Server:  Unknown
Address:  10.87.222.194

Non-authoritative answer:
Name:    www.google.com
Addresses:  2001:4860:4025:7700::
```

Exp No: 1
16/7/26

Basic Networking Commands

AIM:

To execute the basic networking commands in the windows and linux environment

Basic Commands:

1) arp -a : ARP is short form of address resolution Protocol. It will show the IP computer along with the IP address and MAC address of your Router

2) hostname: This is the simplest of all TCP/IP commands. It display the computer's hostname (device name)

3) ipConfig /all : Display complete network configuration including router, gateway, DNS, DHCP and type of ethernet

4) NETStat:

Displays active network connections listening port, router table and network statistics. It is useful for checking open port and current connection

5) nS Lookup: lookup DNS information for a domain or IP address. It returns the IP address of the website and DNS server Details.

```
PS C:\Users\hemes> nslookup www.google.com
Server:  UnKnown
Address:  10.87.222.194

Non-authoritative answer:
Name:    www.google.com
Addresses:  2001:4860:482b:7700::
          2001:4860:482d:7700::
          2001:4860:482a:7700::
          2001:4860:482c:7700::
          2001:4860:4828:7700::
          2001:4860:4826:7700::
          2001:4860:4827:7700::
          2001:4860:4829:7700::
          142.251.157.119
          142.251.150.119
          142.251.155.119
          142.251.156.119
          142.251.152.119
          142.251.154.119
          142.251.153.119
          142.251.151.119

PS C:\Users\hemes> pathping google.com

Tracing route to google.com [142.251.43.46]
over a maximum of 30 hops:
  0  HemeshKumar31 [10.87.222.52]
  1  10.87.222.194
  2  192.0.0.1
  3     *        *        *
Computing statistics for 50 seconds...
PS C:\Users\hemes> ping localhost ping 8.8.8.8
Bad parameter ping.
PS C:\Users\hemes> ping localhost

Pinging HemeshKumar31 [::1] with 32 bytes of data:
Reply from ::1: time<1ms
Reply from ::1: time<1ms
Reply from ::1: time<1ms
Reply from ::1: time<1ms
```

```
PS C:\Users\hemes> ping localhost

Pinging HemeshKumar31 [::1] with 32 bytes of data:
Reply from ::1: time<1ms
Reply from ::1: time<1ms
Reply from ::1: time<1ms
Reply from ::1: time<1ms

Ping statistics for ::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
PS C:\Users\hemes> route print
===========================================================================
Interface List
  8...08 00 27 00 00 08 ......VirtualBox Host-Only Ethernet Adapter
 20...00 ff ab 66 7f 58 ......TAP-Windows Adapter V9
  4...c2 35 32 6d bf e1 ......Microsoft Wi-Fi Direct Virtual Adapter
 17...c6 35 32 6d bf e1 ......Microsoft Wi-Fi Direct Virtual Adapter #2
  3...c0 35 32 6d bf e1 ......Realtek RTL8852BE WiFi 6 802.11ax PCIe Adapter
  1...........................Software Loopback Interface 1
===========================================================================

IPv4 Route Table
===========================================================================
Active Routes:
Network Destination        Netmask          Gateway       Interface  Metric
          0.0.0.0          0.0.0.0    10.87.222.194    10.87.222.52     35
       10.87.222.0    255.255.255.0         On-link     10.87.222.52    291
      10.87.222.52  255.255.255.255         On-link     10.87.222.52    291
     10.87.222.255  255.255.255.255         On-link     10.87.222.52    291
        127.0.0.0        255.0.0.0         On-link         127.0.0.1    331
        127.0.0.1  255.255.255.255         On-link         127.0.0.1    331
  127.255.255.255  255.255.255.255         On-link         127.0.0.1    331
    192.168.56.0    255.255.255.0         On-link     192.168.56.1    281
    192.168.56.1  255.255.255.255         On-link     192.168.56.1    281
  192.168.56.255  255.255.255.255         On-link     192.168.56.1    281
        224.0.0.0        240.0.0.0         On-link         127.0.0.1    331
        224.0.0.0        240.0.0.0         On-link     192.168.56.1    281
        224.0.0.0        240.0.0.0         On-link     10.87.222.52    291
  255.255.255.255  255.255.255.255         On-link         127.0.0.1    331
```

7) Path Ping :

Combines ping and tracert shows the path to a destination and packet lose where packet loss occurs.

8) Ping : Test wheather another device on website is rechable using ICMP Packets shows latency and packet loss.

9) Route : Displays or modifies the IP routing table shows all Route used by the Computer to reach different network

Results :

Thus, the basic networking commands has been successfully executed in the Linux and windows Terminal.